

INDUSTRY WORK EXPERIENCE**NeuroErgonomics Labs, Madison, Wisconsin - Research Scientist | Technical Project Lead****Jan. 2025 – Current**

- Leading cross-functional project teams, coordinating analytical workstreams, timelines, and deliverables across engineers, researchers, and domain experts.
- Contributed to proposal development as well as logistics and resource allocation discussions.
- Providing analytical oversight and mentorship to junior team members, reinforcing structured problem-solving approaches.
- Leading end-to-end research for an NSF-funded immersive training platform supporting high-stress operational environments.
 - Designed simulated law enforcement de-escalation training scenarios in VR using Unity, implemented through Meta Quest 3.
 - Identified gaps in existing emergency response protocols and formulated research hypotheses to guide the development of intelligent virtual training systems using AI-enabled feedback and real-time physiological data.
 - Designed and deployed structured surveys and behavioral experiments using large-scale online participant pools; applied regression modeling, ANOVA, and hypothesis testing to evaluate user attitudes and performance outcomes.
 - Built scalable cloud-based research pipelines (AWS + MTurk) to collect and analyze behavioral interaction data across diverse populations.
 - Designed evaluation of Human-AI interface based on context, literature and subject matter expert driven guidelines.
 - Translated qualitative user research into interface changes that improved task efficiency up to 25%
- Spearheading a multi-modal fatigue detection system to better understand and predict human states of fatigue and stress in individuals with high-stress jobs such as emergency medical service professionals.
 - Collaborated closely with stakeholders to integrate domain-specific insights into system design and usability evaluations.
 - Designed and conducted participant-facing studies to evaluate system usability.
- Developed end-to-end data processing pipelines for physiological, behavioral and contextual data.
 - Created statistical and inferential modeling workflows.
 - Built and validated machine learning models for prediction and forecasting of psychophysiological conditions.
- Translated data-driven insights into actionable system improvements.
- Applied Jobs-to-be-Done frameworks to understand how emergency responders define success, effort, and decision confidence within high-stakes environments.
- Findings informed design iterations, influenced feature prioritization, go/no-go decisions, drove roadmap shifts and were incorporated into subsequent system updates.

Purdue University, Biomedical Engineering, West Lafayette, Indiana – Researcher (Doctoral Studies)**Jan. 2020 – Dec 2024**

- Designed and validated a non-invasive AI-powered tremor diagnostic system using wearable IMUs, improving detection sensitivity beyond clinical gold standards and reducing inter-physician variability.
- Engineered a data-driven tremor signal-based tremor evaluation metric which has shown greater resolution and better sensitivity than the current clinical gold standard in detecting the changes in tremor severity.
- Conducted in-clinic and remote human-subject studies with Parkinson's patients and neurologists, handling sensitive health data in compliance with IRB, HIPAA, and IU Health Medical Research Board guidelines.
- Led in-depth clinician interviews and workflow analyses to identify diagnostic variability and friction points in severity assessment.
- Collaborated with Parkinson's Foundation Center of Excellence and Indiana School of Medicine's Department of Neurology to align the diagnostic system with the current gold standard of clinical measures.
- Conducted structured interviews and iterative feedback sessions with neurologists and clinical researchers to refine diagnostic metrics to improve usability and clinical adoption.
- Achieved 9% improvement in diagnostic accuracy relative to current sensor-based systems through redesigned signal processing pipelines and clinician-informed workflow integration.
- Published and presented in several highly reputed journals and conferences such as Nature and Society for Neuroscience.

US Medical Innovations, Takoma Park, Maryland – Project Lead**Sept. 2016 – Dec. 2019**

- Led cross-functional development of an \$8M surgical robotic platform from concept through verification.
- Conducted workflow and task analyses to define user requirements and identify interaction failure points in high-stakes clinical environments.
- Designed and executed formative usability evaluations with surgeons, assessing efficiency, cognitive load, and interface clarity.
- Synthesized qualitative observations and user feedback into actionable design improvements and system-level mitigations.
- Contributed to multiple patented innovations in electromechanical system design.

Tupelo Brittany, Remote – Equity Consultant Intern**Jan. 2024 – June 2024**

- Delivered data-driven strategy engagements for biotech and pharmaceutical clients, synthesizing scientific literature, company filings (Form 10-K, investor reports), and market data into evidence-based recommendations for investment and portfolio decisions.
- Conducted sector and pipeline analyses connecting therapeutic innovation to competitive dynamics, valuation risk, and R&D investment trends; produced executive-ready summaries and visual presentations for senior stakeholders.

EDUCATION AND CERTIFICATES**Purdue University, West Lafayette, IN: Doctor of Philosophy in Biomedical Engineering****2020 – 2024****Purdue University, West Lafayette, IN: Master of Science in Mechanical Engineering****2016****Purdue University, West Lafayette, IN: Bachelor of Science in Mechanical Engineering****2014****TECHNICAL SKILLS**

- Mixed-Methods Research:** Field research, contextual inquiry, journey mapping, Jobs-to-be-Done research, persona development, ethnographic interviews, usability testing (lab and field), survey design, generative and evaluative research methods.
- Healthcare Research & Compliance:** Human-subject research, IRB protocols, HIPAA compliance, handling protected health information (PHI), clinician and patient research, sensitive topic moderation.

- **Quantitative & Statistical Analysis:** Experimental design, A/B testing frameworks, regression modeling (linear and logistic), ANOVA, hypothesis testing, power analysis, survey validation, mixed-method synthesis, behavioral data modeling.
- **Programming & Data Analysis Tools:** Python, SQL, MATLAB; statistical modeling, data cleaning, visualization, and large-scale behavioral dataset analysis.
- **Systems & Prototyping Experience:** Unity (VR), wearable systems, multimodal data integration.

TEACHING WORK EXPERIENCE

Purdue University, Biomedical Engineering, West Lafayette, Indiana – Graduate Teaching Assistant **Aug. 2021 – Dec 2024**

- Mentored 400+ engineering students in user-centered medical device design and research methods.
- Awarded Outstanding Instructor Award (2023).

Purdue University, EPICS, West Lafayette, Indiana – Graduate Teaching Assistant **Aug. 2014 – May 2016**

- Mentored 400+ engineering students in electromechanical systems design and fabrication, ethics, and user-centered design.
- Awarded Certificate in Practice of College Teaching from the Center for Instructional Excellence at Purdue University

LEADERSHIP EXPERIENCE

- Co-founded the American India Foundation – Purdue Chapter: organized national level fundraising events to support social causes in India.
- Led a cross-functional team at US Medical Innovations, overseeing project goals and deadlines, technical R&D, and budget.
- Co-founded SPREE Technologies, a digital in-video advertisement startup.

RELEVANT RESEARCH PUBLICATIONS

- **Shanghavi, A. A.**, Jayasuriya, J., Cai, Y., & Mehta, R. K. (2025). Pulse Rate Variability From Wearable Wristwatches as a Surrogate for Heart Rate Variability? We Think Not Yet. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 0(0).
- Chen, W., Ruan, W., Pater, K., **Shanghavi, A.**, Mohanty, R., & Mehta, R. (2025, October 13–17). Human factors evaluation of Gen-AI based NPC interactions in public safety training. ASPIRE 2025: 69th International Annual Meeting of the Human Factors and Ergonomics Society, Chicago, IL.
- **A Shanghavi**, D Larranaga, R Patil, E M. Frazier, S Ambike, B S. Duerstock and A B. Sereno. A machine-learning method isolating changes in wrist kinematics that identify age-related changes in arm movement. Scientific Reports 14, 9765 (2024). <https://doi.org/10.1038/s41598-024-60286-1>
- **A. Shanghavi**, R. Patil, J. Stad, G. Maffei, B. S. Duerstock, A. B. Sereno. Power Spectral Density Analysis of Kinematic Variables Reveals Age-related Differences in Arm Motion. Program No. PST151.13/BB14. 2023 Neuroscience Meeting Planner. Washington, D.C.: Society for Neuroscience, 2023. Online
- Sakaleros, J., Shamloo, F., **Shanghavi, A.**, & Sereno, A. (2021). Detection of Parkinsonian Motor Symptoms Using Non-Invasive Motion Sensing Technology. Proceedings of IMPRS, 4(1).
- Andersen, D.; Popescu, V.; Cabrera, M.; **Shanghavi, A.**; Gomez, G.; Marley, S.; Mullis, B. & Wachs, J. (2016). Medical Telementoring Using an Augmented Reality Transparent Display. Surgery, 159(6), 1646 – 1653.
- Andersen, D.; Popescu, V.; Cabrera, M.; **Shanghavi, A.**; Gomez, G.; Marley, S.; Mullis, B. & Wachs, J. Avoiding Focus Shifts in Surgical Telementoring Using an Augmented Reality Transparent Display. NextMed/MMVR22 Proceedings, 2016.
- Dan Andersen, Voicu Popescu, Maria Eugenia Cabrera, **Aditya Shanghavi**, Brian Mullis, Sherri Marley, Gerardo Gomez, Juan P. Wachs; An Augmented Reality-Based Approach for Surgical Telementoring in Austere Environments, Military Medicine, Volume 182, Issue suppl_1, 1 March 2017, Pages 310–315

PATENTS

- Canady, J., O'connell, M., Yan, F., Liu, Z., & **Shanghavi, A.** (2020). U.S. Patent Application No. 29/644,182.
- Canady, J., Canady, C., Zhuang, T., Yan, F., **Shanghavi, A.**, Liu, Z., ... & Pierce, T. (2022). U.S. Patent No. 11,224,487. Washington, DC: U.S. Patent and Trademark Office.
- Canady, J., O'connell, M., Yan, F., Liu, Z., & **Shanghavi, A.** (2020). U.S. Patent Application No. 29/644,181.
- Canady, J., Yan, F., Zhuang, T., **Shanghavi, A.**, Ray, L., Sumanasena, B., & Sokolovski, E. (2023). U.S. Patent Application No. 18/126,799.
- Canady, J., Yan, F., Zhuang, T., **Shanghavi, A.**, Ray, L., Sumanasena, B., & Sokolovski, E. (2023). U.S. Patent No. 11,612,425. Washington, DC: U.S. Patent and Trademark Office.